

Summary

Senior Applied ML & Systems Engineer with 7+ years of technical experience developing scalable machine learning infrastructure, previously in academic/industrial research and currently solving complex problems in AI and agentic privacy. Passionate about building distributed ML serving platforms, automated evaluation testing harnesses, and out-of-model guardrails for ML security. Experienced in productionizing Retrieval-Augmented Generation (RAG) pipelines, LLM-as-a-Judge verification pipelines, and high-throughput bulk inference workflows.

Work Experience

Google

Bangalore, India

Software Engineer – Applied ML & Privacy

Jun. 2024 - Present

- **Production RAG Systems Architecture:** Scoped, architected, and deployed an end-to-end production RAG service providing real-time compliance query interpretation. Reduced operational compute costs by **80%** without degrading accuracy by replacing Majority Voting with a One-Shot Chain of Thought (CoT) reasoning engine, deploying backend via gRPC microservices and standardized agentic APIs.
- **Evaluation Harnesses & Grounding Optimization:** Designed **Semantic Header Injection** and **Deterministic Metadata Extraction** algorithms to resolve context dilution and eliminate citation hallucinations, achieving **98.5% Factual Grounding**. Built automated **LLM-as-a-Judge** verification harnesses to benchmark model migration and resolve recall regressions across frontier LLM releases.
- **Distributed Prompt Safety & Gateway Defenses:** Authored engineering guardrails and reusable developer tooling for automated code-review pipelines to flag prompt injection vulnerabilities and guide architectural mitigations (**Sandwich Defense, Instruction Defense**) with integrations to ML security classifiers.

Google DeepMind & Google Research India

Bangalore, India

Pre-Doctoral Researcher & ML Engineer

Jul. 2022 - Jun. 2024

- **Factual Text Restylization & LoRA:** Developed factual intermediate representations for text-style transfer using LLMs to preserve semantic truth across large corpora. Conducted supervised fine-tuning and parameter-efficient adaptation (**LoRA**) on base foundational models.
- **High-Throughput Bulk Inference Pipelines:** Engineered distributed, high-throughput bulk inference pipelines (JAX, Apache Beam) integrated with automated downstream metric collection.
- **Multimodal Conversational Interfaces (VLMs):** Designed and built an interactive conversational interface over complex data visualizations integrating Vision-Language Models (VLMs) and LLMs. Built core orchestration logic and multimodal data parsing pipelines with C++/Python test suites.

Google through Optimum Infosystems

Bangalore, India

Data Commons Associate

Aug. 2021 - Jun. 2022

- **Knowledge Graph Schema & Scalable ETL Workflows:** Expanded Google's open statistical knowledge graph (datacommons.org) by modeling novel statistical schemas, building high-throughput Python/Pandas data ingestion workflows, and committing production open-source pull requests.

Centre for Networked Intelligence, Indian Institute of Science

Bangalore, India

Technical Associate

Dec. 2019 - Jul. 2021

- Co-developed software architecture of COVID-19 epidemiological agent-based simulation models (*ABMaaS*, *Serosurvey Optimization*).

Education

Delft University of Technology (TU Delft)

Delft, the Netherlands

M.S. in Computer Science

Sep. 2017 - Aug. 2019

- Thesis: "[A Human-Machine Approach to Preserve Privacy in Image Analysis Crowdsourcing Tasks](#)".

Amrita Vishwa Vidyapeetham (Amrita University)

Coimbatore, India

B.Tech. in Computer Science and Engineering

Aug. 2012 - May. 2016

Highlights

- Received multiple Google internal awards (Peer and Spot bonuses) for outstanding technical delivery, systems design, and cross-functional leadership.
- Received the *IBM Benelux Excellence Award* and placed as Runner-up for *Best Innovation Impact* at the IBM Extreme Blue European Expo (2018).

Selected Publications

- A Talekar, **Shriram, S**, N Vaidhiyan, G Aggarwal, J Chen, S Venkatramanan, L Wang, A Adiga, A Sadilek, A Tendulkar, M Marathe, R Sundaresan, and M Tambe. [Cohorting to Isolate Asymptomatic Spreaders: An Agent-Based Simulation Study on the Mumbai Suburban Railway](#). *Proceedings of the 20th International Conference on Autonomous Agents and MultiAgent Systems*, pages 1680–1682, 2021.
- S Agrawal, S Bhandari, A Bhattacharjee, A Deo, N M Dixit, P Harsha, S Juneja, P Kesarwani, A K Swamy, P Patil, et al. City-scale agent-based simulators for the study of non-pharmaceutical interventions in the context of the covid-19 epidemic. *Journal of the Indian Institute of Science*, pages 1–39, 2020.
- **Sharad, S***, P B Sivakumar, and V Anantha Narayanan. [The smart bus for a smart city—A real-time implementation](#). In *2016 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, pages 1–6. IEEE, 2016.
- P Parag, P Patil, **S Shriram**, R Sundaresan, H Tyagi, and N K Vaidhiyan. [Antidote for CoVID-19 pandemic induced surge in online learning](#). In *Proceedings of the ACM SIGCOMM Online Networking Education Discussion Workshop*, 2020.
- S Athreya, G R Babu, A Iyer, M M B S, N Rathod, **Shriram, S**, R Sundaresan, N Vaidhiyan, and S Yasodharan. [COVID-19: Optimal Design of Serosurveys for Disease Burden Estimation](#). *Sankhya B*, pages 1–23, 2020.
- **Sharad, S***, P B Sivakumar, and V Ananthanarayanan. [An automated system to mitigate loss of life at unmanned level crossings](#). *Procedia computer science*, 92:404–409, 2016

Skills

Applied ML & GenAI	Agentic Workflows, Retrieval-Augmented Generation, Prompt Engineering, LLM-as-Judge Evaluation Methods, Multimodal Systems
ML/NLP Frameworks	TensorFlow, Hugging Face, PyTorch, NumPy, LoRA
Programming Languages	Python, C++, JavaScript, Java
Data & Infrastructure	Pandas, Apache Beam, SQL, Scipy